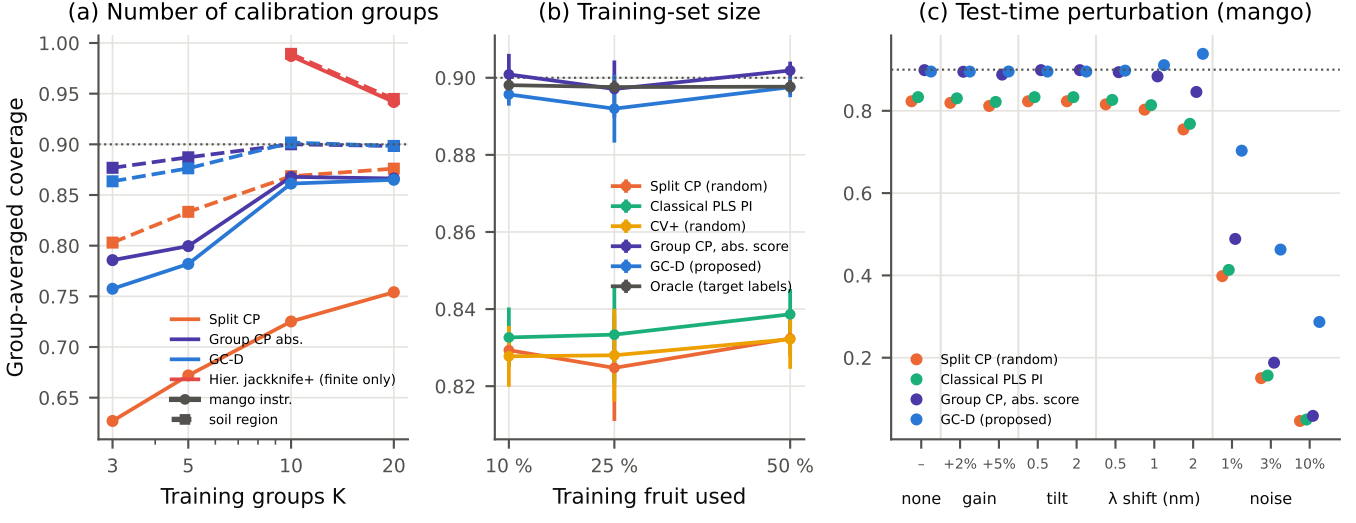
(a) Number of calibration groups
Group-averaged coverage
1.00
0.95
0.90
0.85
0.80
0.75
0.70
0.65
3
5
10
20
Training groups K
Split CP
Group CP abs.
GC-D
Hier. jackknife+ (finite only)
mango instr.
soil region
(b) Training-set size
0.90
0.88
0.86
0.84
0.82
10 %
25 %
50 %
Training fruit used
Split CP (random)
Classical PLS PI
CV+ (random)
Group CP, abs. score
GC-D (proposed)
Oracle (target labels)
(c) Test-time perturbation (mango)
0.8
0.6
0.4
0.2
Split CP (random)
Classical PLS PI
Group CP, abs. score
GC-D (proposed)
–
+2%
+5%
0.5
2
0.5
1
2
1%
3%
10%
none
gain
tilt
λ shift (nm)
noise